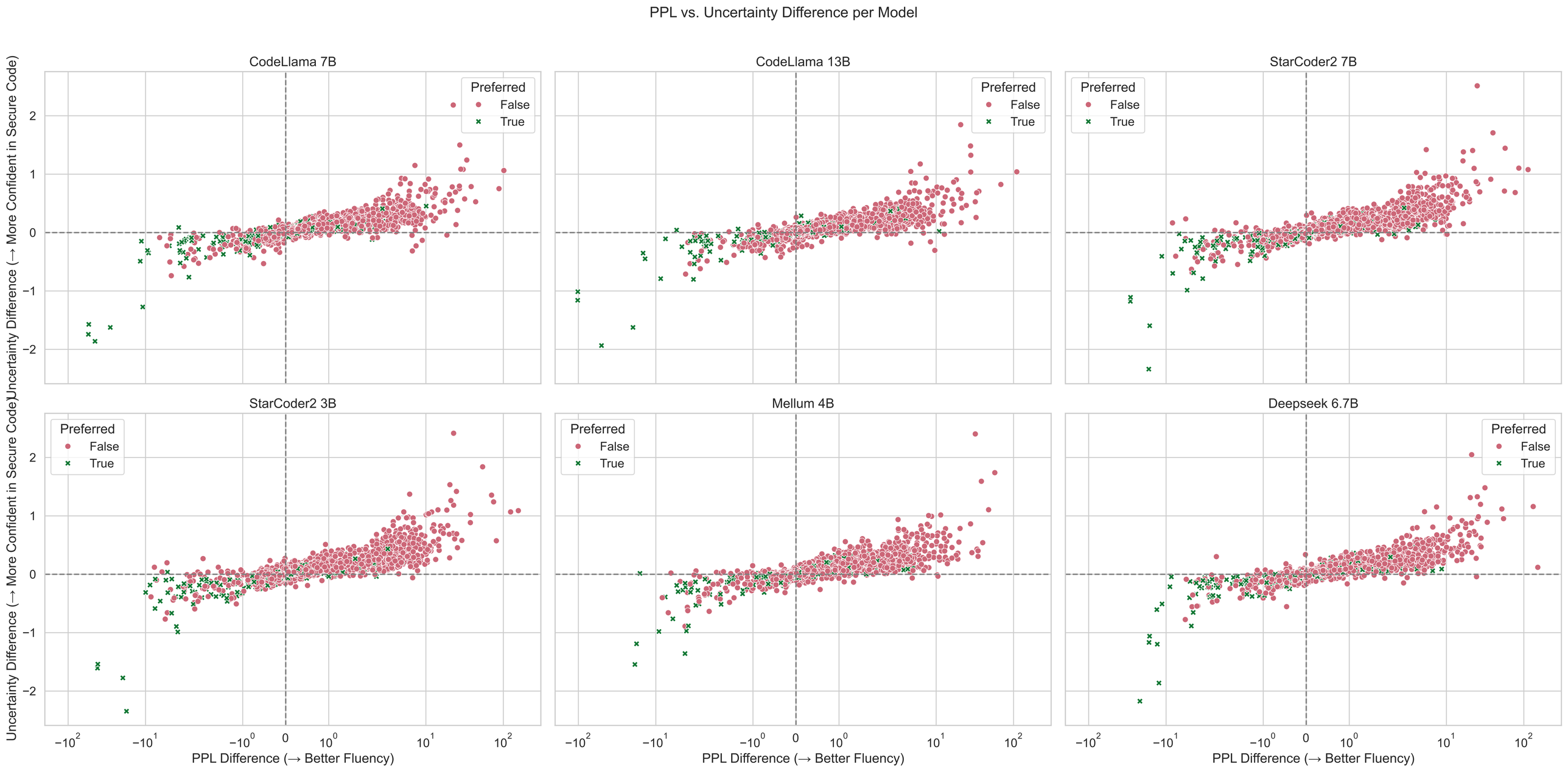
PPL vs. Uncertainty Difference per Model
CodeLlama 7B
CodeLlama 13B
StarCoder2 7B
StarCoder2 3B
Mellum 4B
Deepseek 6.7B
Preferred
False
True
Uncertainty Difference (→ More Confident in Secure Code)
PPL Difference (→ Better Fluency)
2
1
0
−1
−2
-10^2
-10^1
-10^0
0
10^0
10^1
10^2